# A Modular Computational Framework for Section 2 Voting Rights Act Analysis

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## Abstract

A modular computational framework for §2 Voting Rights Act is introduced which integrates geospatial data engineering, Bayesian ecological inference, and ensemble generation. Prior work has developed these components individually, existing approaches lack a unified pipeline that propagates uncertainty from voter behavior estimation through plan-level evaluation. A probabilistic notion of minority electoral opportunity is formalized. A case study of Texas congressional districts illustrates how the framework can be used to assess whether an enacted plan systematically dilutes minority voting power.

## Introduction

### Gerrymandering

Gerrymandering is a tactic that has been interwoven with United States politics since at least 1812, which is when the term was first coined. In simplest terms, gerrymandering can be defined as the manipulation of election district maps to either favor or disfavor a group of people (i.e. a political party, a racial/ethnic group, a socioeconomic class, etc).

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## Negative Racial Gerrymandering

Negative racial gerrymandering dilutes minority voting power by engineering districts in which minority ballots systematically matter less. Although minority voters retain the formal right to vote, they lack proportional influence.

This distortion produces legislatures that are less representative—often whiter and more ideologically extreme than the electorate—and weakens elected officials' incentives to respond to minority interests.

By breaking the link between voting and representation, negative racial gerrymandering undermines democratic legitimacy, erodes trust, and contributes to lower turnout and political disengagement.

## The Legality of Negative Racial Gerrymandering

Negative racial gerrymandering is illegal in the United States by virtue of the Equal Protection Clause of the Fourteenth Amendment and Section 2 of the Voting Rights Act.

The Equal Protection Clause of the Fourteenth Amendment forbids denying anyone of "equal protections of the law" [United States Congress, 1868].

Section 2 of the Voting Rights Act of 1965 precludes states and political entities from wielding any "standard, practice, or procedure" to curtail the voting rights of any U.S. citizen based on race [United States Congress, 1965]. In the 1986 case of Thornburg v. Gingles, it was ruled that a North Carolina redistricting plan was in violation of Section 2 of the Voting Rights Act for engaging in negative racial gerrymandering practices - more specifically, of diluting the votes of black citizens [Supreme Court of the United States, 1986]. This landmark Supreme Court case decided that minority voters can legally challenge negative racial gerrymandering practices in redistricting plans on the grounds of violating Section 2 of the Voting Rights Act, but that the burden of proof is on them to demonstrate that their capacity to elect their preferred candidates is undermined [Supreme Court of the United States, 1986].

In the Thornburg v. Gingles case, the Supreme Court established a framework for minority voters to be able to prove negative racial gerrymandering in violation of Section 2 of the Voting Rights Act in a court of law [Supreme Court of the United States, 1986]. Under this framework, plantiffs must be able to prove the following: (1) a "geographically insular" minority group makes up the majority population of a district, (2) the minority group is "politically cohesive", and (3) the "bloc voting majority" is usually able to defeat the minority group's preferred candidates [Supreme Court of the United States, 1986]. The second and third preconditions are stated by the U.S. Supreme Court as concerning "racially polarized voting" [Supreme Court of the United States, 2023].

The U.S. Supreme Court refers to this framework as "the Gingles framework" [Supreme Court of the United States, 2023]. For the last 40 years, claims of negative racial gerrymandering in violation of Section 2 of the Voting Rights Act have been assessed via the Gingles framework [Supreme Court of the United States, 2023]. In the 2023 Allen v. Milligan decision, the U.S. Supreme Court reaffirmed the validity of the Gingles framework [Supreme Court of the United States, 2023].

# Related Work

## Redistricting Data Preparation

### Constructing Racial Categories

In this paper, the Data and Democracy Lab's conventions for constructing demographic categories are adopted.

Their convention for categorizing racial groups involves adopting these group names in the ensuing order: Black, Hispanic, Asian, Other,and White [Donnay, M. and Duchin, M. and Rock, P., n.d.].

It is emphasized that anyone who identifies as black should be grouped into the black population [Donnay, M. and Duchin, M. and Rock, P., n.d.]. In other words, the black population should encompass those who are "Any-Part Black" or "Black Alone or in Combination", as opposed to just "Black Alone" or "Single-Race Black" [Donnay, M. and Duchin, M. and Rock, P., n.d.].

Then, of the individuals left, the Hispanic population must encompass all individuals who identify as Hispanic in the same way the Black population does [Donnay, M. and Duchin, M. and Rock, P., n.d.]. Next, of those left, the Asian population should capture all who identify as Asian or Native Hawaiin Pacific Islander, so as to represent an AAPI category [Donnay, M. and Duchin, M. and Rock, P., n.d.]. Finally, the same logic is used to construct the AMIN, Other, and White categories [Donnay, M. and Duchin, M. and Rock, P., n.d.].

Thereafter, source data:pl94 in Section sec:datasources is leveraged to generate voting age population as sums of census variables [Donnay, M. and Duchin, M. and Rock, P., n.d.].

Subsequently, Source **??** in §@ref(datasources) is analyzed. The problem is, CVAP data from the ACS must be analyzed at the census block level because the base unit of the Decennial Census PL 94-171 data is the census block. In the primary tables, ACS is reported at the census tract level, which presents a problem because that is a significantly higher level of aggregation than the census block [Donnay, M. and Duchin, M. and Rock, P., n.d.].

Fortunately, the Data and Democracy Lab has introduced a solution to this problem called "discounting" [Donnay, M. and Duchin, M. and Rock, P., n.d.]. Discounting refers to the usage of the tract-level ACS data to estimate the citizenship rates of racial groups, then applying the discount to the voting age population [Donnay, M. and Duchin, M. and Rock, P., n.d.].

An example of "discounting" provided by the Data and Democracy Lab is depicted in Equation~**??**; in this example, block X belongs to tract Y in state Z and BCVAP is estimated by taking the citizenship rate for the group in the tract and applying it to the voting ager population in the block [Donnay, M. and Duchin, M. and Rock, P., n.d.].

$$\frac{BCVAP_{ACS}(tract\,Y)}{BVAP_{ACS}(tract\,Y)} \times BVAP_{PL}(block\,X) = BCVAP(block\,X)$$

If the denominator in an equation like Equation~**??** is too low, then the Data and Democracy Lab proposes a solution like that in Equation~**??**. This involves swapping in the citizenship rate given that the tract did not allow for a meaningful estimate [Donnay, M. and Duchin, M. and Rock, P., n.d.].

$$\frac{BCVAP_{ACS}(state\,Z)}{BVAP_{ACS}(state\,Z)} \times BVAP_{PL}(block\,X) = BCVAP(block\,X)$$

The Data and Democracy Lab acknowledges that this is a "crude fix", but states that they intend to release a "more sophisticated" Bayesian method to address this in the future [Donnay, M. and Duchin, M. and Rock, P., n.d.].

## Ecological Inference

To reiterate what is stated in §@ref(legality), in order to prove a violation of §2 of the Voting Rights Act under the Gingles framework,plaintiffs must prove that the majority votes as a bloc to defeat minority-preferred candidates [Supreme Court of the United States, 1986]. Hence, any framework performing §2 Voting Rights Act analysis must encompass identification of minority-preferred political candidates [Becker et al., 2021].

In state election return datasets, vote totals for political candidates are not computed by race, meaning that direct computation of minority-preferred candidates is impossible [Becker et al., 2021]. This is where ecological inference - the de-facto solution for determining racially polarized voting - comes in. After all,

ecological inference is widely regarded as "the best and often only hope of making progress" when it comes to extracting clues about individual behavior from group or aggregate-level data [King et al., 2004].

Ecological inference (and ecological regression, its simpler predecessor) is considered a "statistical mainstay" in §2 Voting Rights Act litigation [Freedman, 1999]. U.S. Supreme Court redistricting cases in which lower-court findings relying on ecological inference were reviewed and accepted without methodological objection include Allen v. Milligan [Supreme Court of the United States, 2023], Alexander v. State Carolina State Conference of the NAACP [Supreme Court of the United States, 2024], Cooper v. Harris [Supreme Court of the United States, 2017a], Bethune-Hill v. Virginia State Board of Elections [Supreme Court of the United States, 2017b], and Abbott v. Perez [of the United States, 2018].

It should be noted, however, that in the same way all statistical inference techniques with missing data give way to uncertainty, ecological inference produces "imperfect" and "uncertain" answers [Becker et al., 2021]. Hence, it is essential to estimate the error produced by ecological inference techniques and observe how it compounds or cancels out in high-level conclusions [Becker et al., 2021].So, in addition to identifying the minority-preferred candidates, one must specify the confidence measures that the identifications are correct [Becker et al., 2021]. To check for minority control of a district and identify preferred candidates for newly proposed districts, the statewide and precinct-level vote estimates by race should be leveraged [Becker et al., 2021].

Becker et al. [Becker et al., 2021] implement a version of King's ecological inference, specifically the `ei.mD.bayes` function from eiPack which is based on the Bayesian hierarchical Dirichlet model for $R \times C$ tables proposed by King et al. in 1999 [King et al., 1999].

For each election, EI is run at the statewide level using precinct-level input tables [Becker et al., 2021]. The inputs for each precinct are the row and column sums for the $R \times C$ table of vote counts. The row sums correspond to the precinct's estimated number of adult citizens in each racial group. The column sums are the precinct's vote totals for each candidate [Becker et al., 2021].

Each ecological inference run generates a large random sample of estimated vote counts; in order to obtain the statewide estimates, these values are summed across the entire state [Becker et al., 2021]. For each racial group, the candidate with the highest average estimated vote total for an election is the preferred candidate [Becker et al., 2021]. In order to obtain a measure of confidence for the a candidate was correctly identified as the preferred candidate, repeated draws for the ecological inference distribution are obtained and the frequency by which the preferred candidate receives the most votes is recorded [Becker et al., 2021].

# The Framework

## Data Engineering Layer

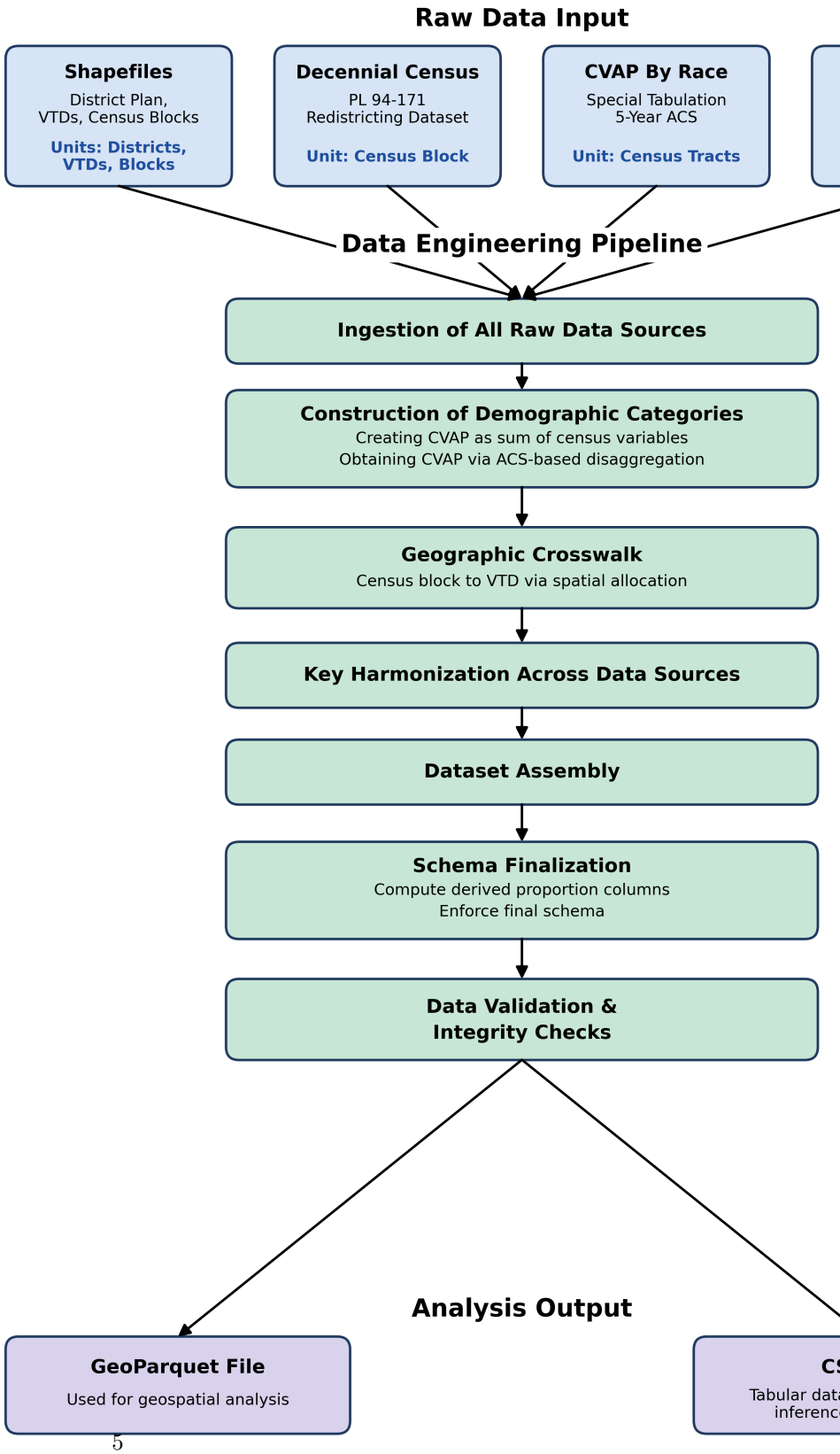




**Figure: Data Engineering Pipeline**

**Data Sources**

1. State District Plan Shapefile
2. Decennial Census PL 94-171 Redistricting Dataset
3. Citizen Voting Age Population by Race and Ethnicity in a Special Tabulation from the ACS 5-Year Estimates
4. State Census Blocks Shapefile (TIGER/LINE)
5. State Voting Tabulation District Shapefile
6. State VTDs Election Data

# Case Study: Texas Congressional Districts

## Data Engineering Layer Implementation

**Table : Texas Demographics**

| Racial Group | Share of Total Population | Share of VAP | Share of CVAP |
|---|---|---|---|
| Hispanic | 34.7% | 35.6% | 32.6% |
| Black | 13.5% | 13.0% | 12.9% |
| White | 46.2% | 43.2% | 48.9% |
| Asian | 5.0% | 6.1% | 4.8% |
| American Indian | 0.7% | 1.3% | 0.7% |

**Table : Texas Statewide Elections Used**

| Office | Stage | Year |
|---|---|---|
| President | General | 2024 |
| President | Primary | 2024 |
| U.S. Senate | General | 2024 |
| U.S. Senate | Primary | 2024 |
| Railroad Commissioner | General | 2024 |
| Railroad Commissioner | Primary | 2024 |

## Ecological Inference Layer Implementation

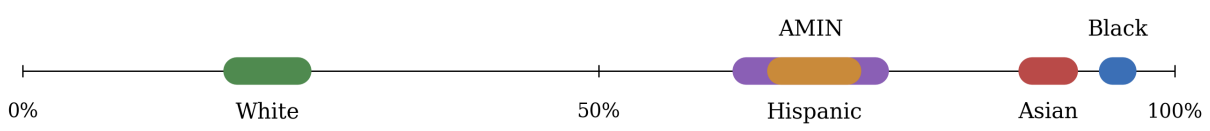


Figure 1: The highest and lowest EI point estimates for each racial group's support of the 3 Democratic nominees in statewide general elections: White (19–24%), AMIN (63–74%), Hispanic (66–72%), Asian (88–90%), and Black (95–96%).

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